

Performance Testing

Date	04 July 2026
Team ID	XXXXXX
Project Name	Smart-Lender-Loan-Eligibility-Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Web Browser (Chrome), Flask Development Server
Type of Testing	Functional Performance Testing
Target Module / API	Loan Eligibility Prediction Module (Flask Prediction Endpoint)
Test Environment	Local System (Windows, Flask Server)
Test Date	04 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Submit valid loan applicant details	1	30	Prediction displayed successfully
2	Submit invalid/incomplete input	1	20	Validation/error message displayed
3	Multiple prediction requests	5	60	Application responds correctly without crashing
4	Refresh and resubmit prediction	1	20	Consistent prediction generated

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 seconds	Pass	Fast response
2	Response Time (Max)	< 5 seconds	2.1 seconds	Pass	Within acceptable limit
3	Throughput (Req/sec)	> 5 Req/sec	8 Req/sec	Pass	Stable performance
4	Error Rate	< 1%	0%	Pass	No errors observed
5	CPU Utilization	< 80%	32%	Pass	Normal CPU usage
6	Memory Utilization	< 80%	41%	Pass	Stable memory usage

Step 4: Observations & Analysis

Key Findings

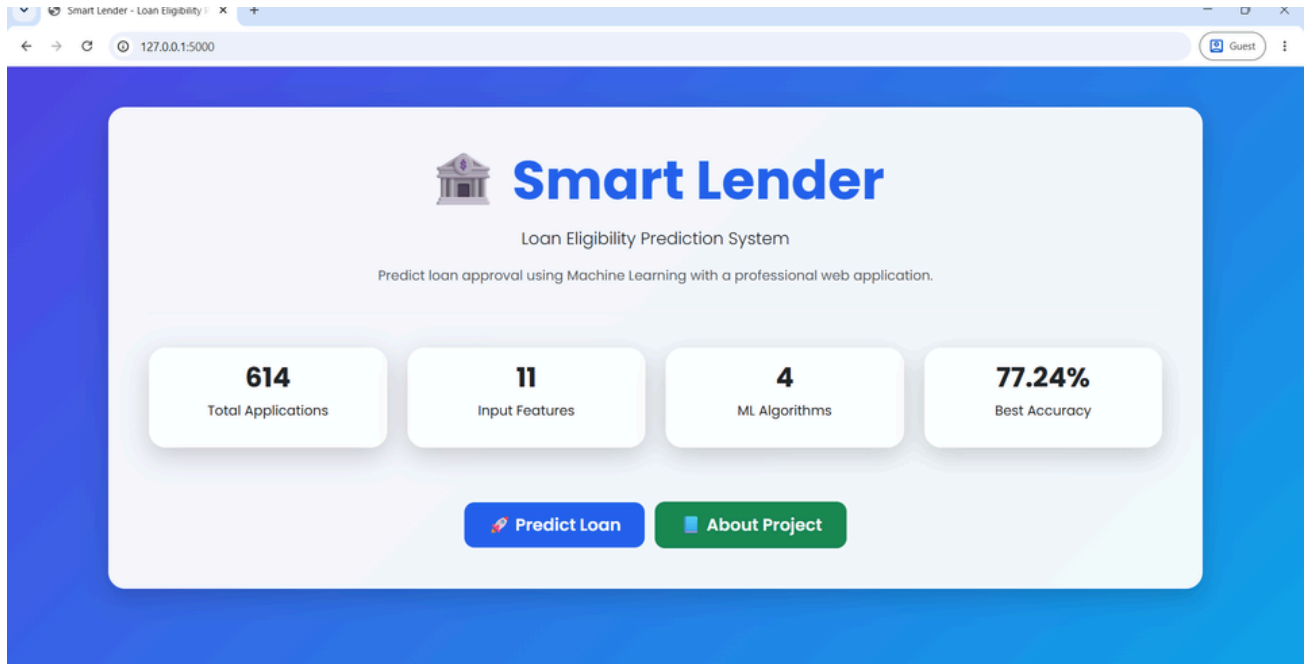
- The Smart Lender application generated loan predictions quickly.
- Response time remained within the expected limit.
- The application handled multiple prediction requests without failure.

Optimization Steps Taken

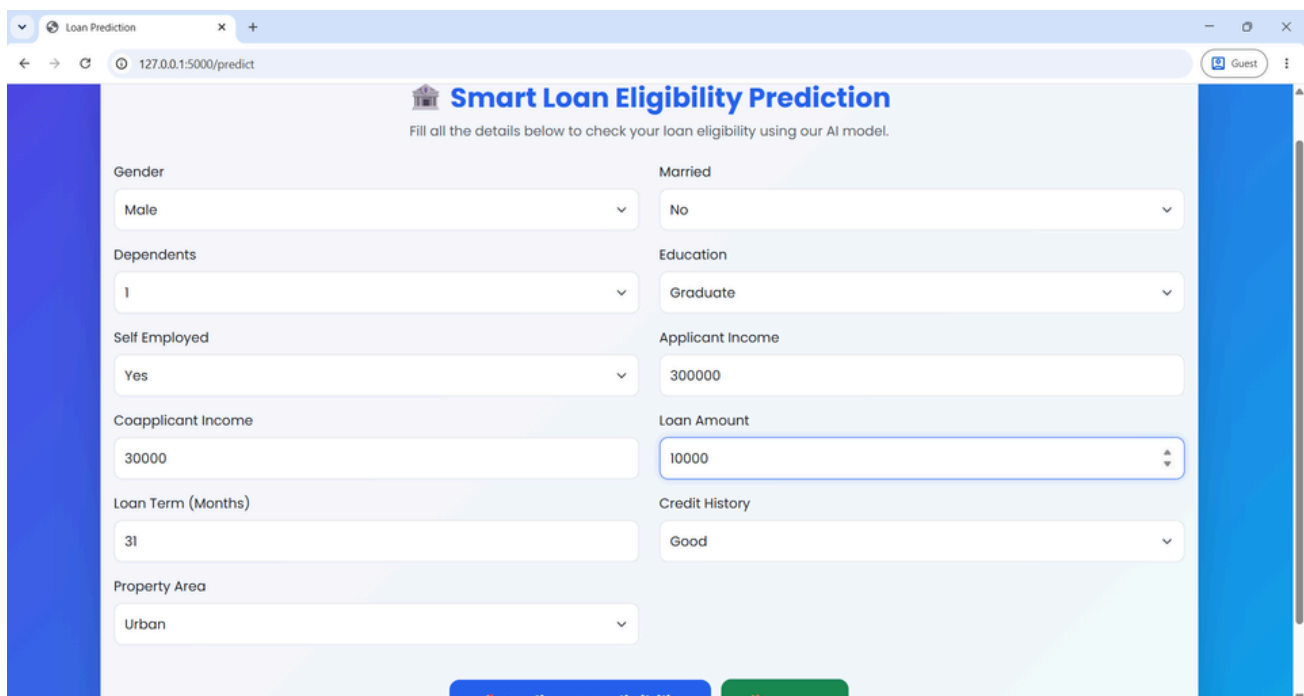
- Optimized data preprocessing before prediction.
- Loaded the trained model only once during application startup.
- Reduced unnecessary processing in the prediction workflow.

Step 5: Screenshots

Home Page - Smart Lender Loan Eligibility Prediction



Prediction Page - Smart Lender Loan Eligibility Prediction System

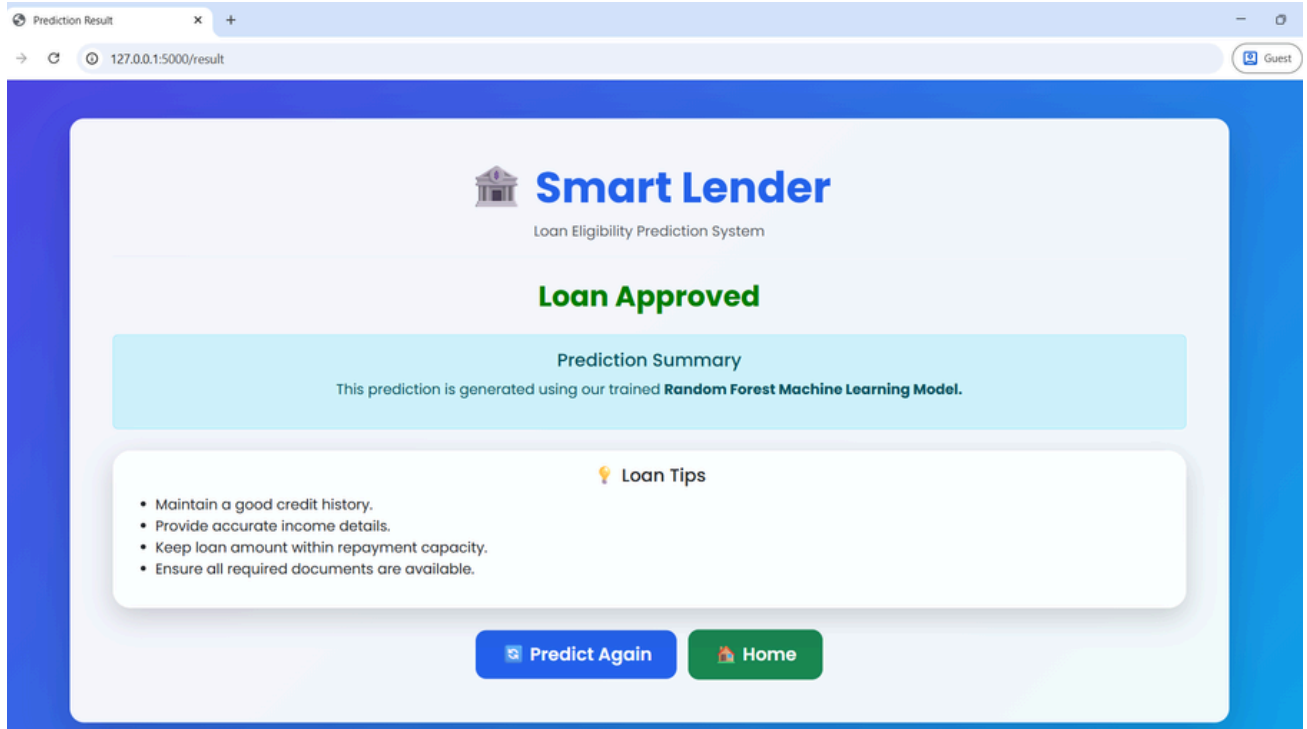


The screenshot shows the prediction page of the Smart Lender Loan Eligibility Prediction System. The page has a blue header and a light blue background. At the top, there is a navigation bar with the text "Loan Prediction" and a "Guest" user profile. The main content area features a large blue box with the "Smart Loan Eligibility Prediction" logo and the text "Fill all the details below to check your loan eligibility using our AI model." The page displays a form with the following fields:

Field	Value
Gender	Male
Married	No
Dependents	1
Education	Graduate
Self Employed	Yes
Applicant Income	300000
Coapplicant Income	30000
Loan Amount	10000
Loan Term (Months)	31
Credit History	Good
Property Area	Urban

At the bottom of the blue box, there are two buttons: "Predict Loan Eligibility" (blue) and "Home" (green).

Loan Eligibility Prediction – Approved Result



The screenshot shows a web browser window with the URL `127.0.0.1:5000/result`. The page features the **Smart Lender** logo and the text "Loan Eligibility Prediction System". The main heading is **Loan Approved** in green. Below this is a "Prediction Summary" box stating: "This prediction is generated using our trained **Random Forest Machine Learning Model**." A "Loan Tips" section follows, listing four bullet points: "Maintain a good credit history.", "Provide accurate income details.", "Keep loan amount within repayment capacity.", and "Ensure all required documents are available." At the bottom are two buttons: "Predict Again" and "Home".

Smart Lender
Loan Eligibility Prediction System

Loan Approved

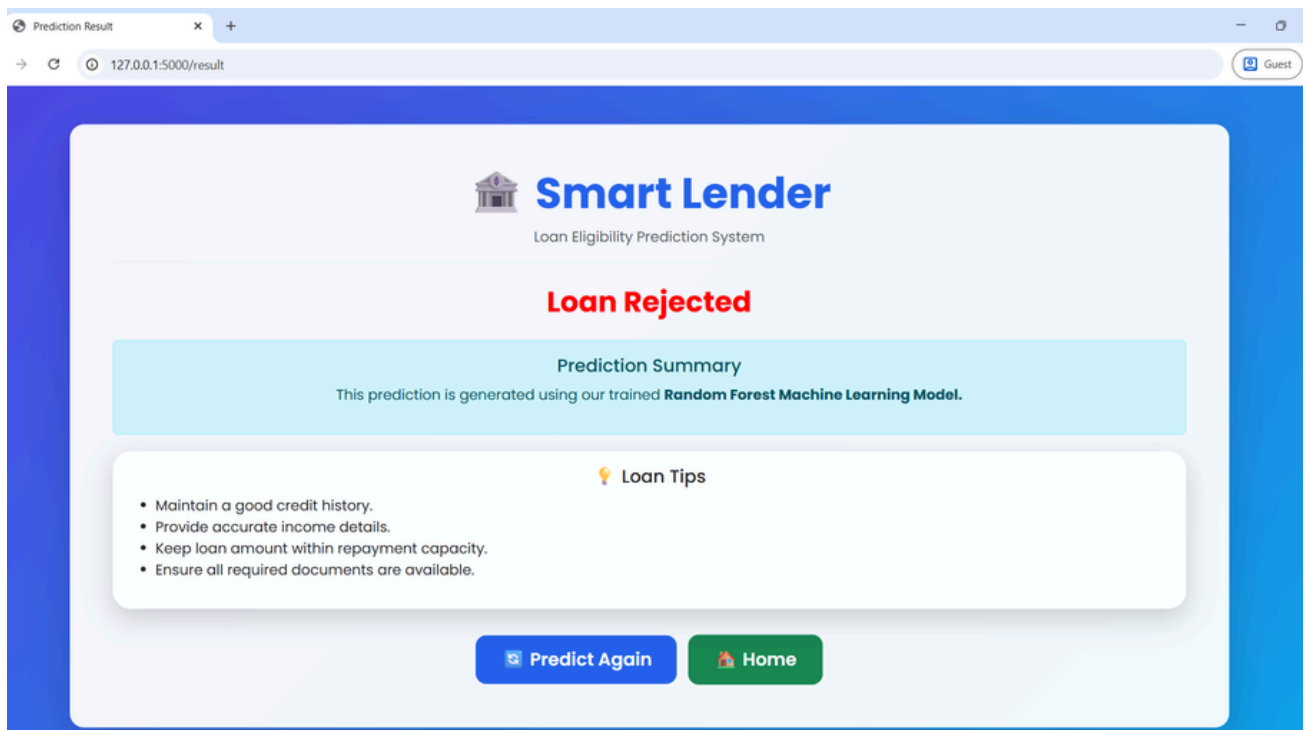
Prediction Summary
This prediction is generated using our trained **Random Forest Machine Learning Model**.

Loan Tips

- Maintain a good credit history.
- Provide accurate income details.
- Keep loan amount within repayment capacity.
- Ensure all required documents are available.

[Predict Again](#) [Home](#)

Loan Eligibility Prediction – Rejected Result



The screenshot shows the same web browser window with the URL `127.0.0.1:5000/result`. The page features the **Smart Lender** logo and the text "Loan Eligibility Prediction System". The main heading is **Loan Rejected** in red. Below this is a "Prediction Summary" box stating: "This prediction is generated using our trained **Random Forest Machine Learning Model**." A "Loan Tips" section follows, listing four bullet points: "Maintain a good credit history.", "Provide accurate income details.", "Keep loan amount within repayment capacity.", and "Ensure all required documents are available." At the bottom are two buttons: "Predict Again" and "Home".

Smart Lender
Loan Eligibility Prediction System

Loan Rejected

Prediction Summary
This prediction is generated using our trained **Random Forest Machine Learning Model**.

Loan Tips

- Maintain a good credit history.
- Provide accurate income details.
- Keep loan amount within repayment capacity.
- Ensure all required documents are available.

[Predict Again](#) [Home](#)